PhysioRisk: an interpretable survival-modeling framework for separating demographic and physiological mortality risk

Gregory S. Allen
Independent researcher, Cuenca, Ecuador
Corresponding author: Gregory S. Allen
Email: allengr220@gmail.com

Abstract

Background: Biomarker-based mortality scores often combine chronological age with physiological measurements, obscuring whether predicted risk reflects demographic mortality structure or physiological deviation. We asked whether mortality risk could be decomposed into a demographic component and a physiological component adjusted for age and sex while preserving prognostic information.

Methods: We analyzed unweighted mortality-linked NHANES 1999–2016 data. DemoRisk was an age- and sex-based Cox predictor trained in 1999–2010. PhysioRisk was derived from development-residualized biomarkers summarized in two physiological axes and applied unchanged to 2011–2016 data for temporal validation. CalibratedTotalRisk recombined the fixed components using development-estimated coefficients. We assessed hazard ratios, Harrell C-index with bootstrap confidence intervals, paired C-index contrasts, proportional-hazards diagnostics, fixed-horizon calibration, IPCW Brier scores, and censoring-robust discrimination. We compared PhysioRisk to PhenoAge NCP, a no-CRP proxy.

Results: Of 92,062 source-cycle participants, 19,829 formed the biomarker-complete cohort: 13,090 for development and 6,739 for temporal validation. PhysioRisk showed low age correlation (r=0.102), C-index 0.663, and HR 1.385 per SD. CalibratedTotalRisk had C-index 0.844, exceeding DemoRisk by 0.018 (95% CI 0.012–0.024), while differing from PhenoAge NCP by −0.0003 (95% CI −0.007 to 0.006). Five- and 8-year predicted mortality showed close overall agreement with observed mortality.

Conclusions: Age- and sex-adjusted physiology retained mortality information and improved discrimination when recombined with demographic risk. Explicit demographic–physiological decomposition preserved useful risk ranking without evidence of superiority over PhenoAge NCP. Findings remain internally validated within NHANES, and external validation is required.

Keywords: mortality prediction; survival analysis; physiological aging; biomarkers; NHANES; risk decomposition

Introduction

All-cause mortality is decidedly structured by chronological age. Prognostic models that combine age with clinical biomarkers can therefore achieve high discrimination even when much of their risk ranking reflects demographic rather than physiological variation. This is not a defect when mortality prediction is the sole objective. It becomes an interpretive problem when mortality-trained scores are also used to describe biological or physiological aging, because a high score may reflect physiological dysfunction, greater chronological age, or some combination of the two (Ferrucci et al. 2018; Klemera and Doubal 2006; Liu et al. 2018; Lu et al. 2019).

Several established approaches address different aspects of this problem. PhenoAge and related mortality-trained aging measures combine age and biomarker information to construct predictors of morbidity and mortality (Liu et al. 2018). Epigenetic mortality predictors such as GrimAge likewise emphasize prognosis, although their molecular inputs and interpretation differ substantially (Lu et al. 2019). Physiological dysregulation methods instead measure multivariate departure from an empirically defined reference state and have shown that aging-related risk can emerge from coordinated changes across physiological systems (Cohen et al. 2013, 2015; Milot et al. 2014). Frailty and deficit-accumulation approaches similarly emphasize heterogeneity among individuals of the same chronological age (Duchateau and Janssen 2007; Rockwood and Mitnitski 2007; Vaupel et al. 1979). These articles suggest an important distinction between predicting total mortality risk and characterizing physiological variation within the demographic structure of that risk.

Chronological age is both biologically informative and statistically dominant. Removing it from a mortality model simply to produce a less age-correlated score would discard important prognostic information. Yet retaining age

within a single aggregate score makes it difficult to determine how much risk reflects the expected mortality gradient associated with age and sex and how much reflects an individual's physiological state relative to others of similar age and sex. These are different questions and need not be answered by the same model component.

PhysioRisk makes this separation explicit. DemoRisk summarizes mortality-risk structure associated with age and sex, whereas PhysioRisk summarizes mortality-associated deviations of clinical biomarkers from values expected conditional on age and sex. The physiological component is constructed only after age- and sex-associated biomarker trends have been removed. The two components can then be examined separately or recombined, using weights estimated in the development cohort, to form CalibratedTotalRisk.

This construction differs from an age-acceleration residual derived from an already assembled biological-age or mortality score. In an acceleration approach, the aggregate score is constructed first and its association with chronological age removed afterward. In PhysioRisk, age and sex are removed from the individual physiological measurements before those measurements are summarized and related to mortality. These operations are not generally equivalent. PhysioRisk therefore asks a different question: whether physiological deviations conditional on age and sex contain mortality information while demographic mortality risk remains separately observable. This distinction does not imply that either approach is intrinsically superior.

Interpretability is thus part of the model architecture rather than a post hoc explanation of a single risk score. The retained PhysioRisk specification organizes the residualized measurements into two transparent physiological axes, HematologicStress and RenalHepaticTurnover. These axes are not intended as complete biological systems or causal mechanisms. They provide a low-dimensional organization of correlated clinical measurements that keeps the physiological contribution to mortality risk inspectable rather than absorbing it into an age-entangled predictor.

We examined whether this decomposition remained useful under internal temporal validation. Specifically, we asked whether age- and sex-adjusted physiological variation retained an association with mortality, whether PhysioRisk contributed prognostic information beyond DemoRisk, whether development-calibrated recombination preserved overall mortality discrimination, and how the resulting composite compared with a Levine-style PhenoAge no-CRP proxy and its age-adjusted residual. We also evaluated uncertainty in discrimination, fixed-horizon calibration, proportional-hazards behavior, censoring-robust performance, missingness, and cohort selection. The objective was not to establish PhysioRisk as a superior biological-age score, but to test whether demographic–physiological risk decomposition provides a useful and interpretable alternative representation of mortality risk. Reporting was guided by TRIPOD (Collins et al. 2015).

Conceptual framework for risk decomposition

In a Cox proportional-hazards model, mortality risk for individual i at time t can be written as

$$h_i(\mathrm{t}) = h_0(\mathrm{t})\,\exp(\eta_i)$$

where $h_0(t)$ is the unspecified baseline hazard and η_i is the individual's covariate-dependent linear predictor (Cox 1972; Kleinbaum and Klein 2012). A useful conceptual distinction is therefore between the baseline hazard, which is common to individuals at a given analysis time, and covariate-dependent differences in relative hazard. In this framework, DemoRisk is not the Cox baseline hazard. It is the portion of the covariate-dependent predictor derived from age and sex.

We write the modeled relative-risk component schematically as

$$\eta_i = \beta_D\, D_i + \beta_P\, P_i$$

where D_i denotes DemoRisk and P_i denotes PhysioRisk. The coefficients β_D and β_P are estimated jointly in the development cohort, and their frozen weighted combination defines CalibratedTotalRisk. The relative numerical scales of the demographic and physiological scores need not be equal. Standardizing one component therefore does not justify adding the two with unit weights.

DemoRisk is derived from a Cox model containing spline-transformed chronological age and sex. PhysioRisk is constructed differently. For each biomarker X_j, the expected value conditional on age and sex is estimated in the development cohort:

$$\mathrm{E}(X_{ij} \mid \mathrm{age}_i, \mathrm{sex}_i) = f_j(\mathrm{age}_i, \mathrm{sex}_i)$$

The residual

$$R_{ij} = X_{ij} - \hat{f}_j(\text{age}_i, \text{sex}_i)$$

represents the participant's deviation from the development-cohort expectation for that biomarker at the same age and sex. The fitted residualization functions and subsequent standardization parameters are then applied unchanged to the temporal-validation cohort.

PhysioRisk is therefore not intended to estimate the total physiological changes that accompany aging. Many genuine biological consequences of aging will, by design, be absorbed into expected age-associated biomarker trajectories and removed from the physiological predictors. The question posed by PhysioRisk is narrower: among people positioned similarly with respect to age and sex, do deviations in measured physiology identify additional mortality heterogeneity?

This distinction also clarifies the relationship to biological-age acceleration measures. If B_i is an aggregate mortality-trained biological-age score, an acceleration measure can be represented schematically as

$$A_i = B_i - \mathrm{E}(B_i \mid \text{age}_i, \text{sex}_i)$$

PhysioRisk instead residualizes the biomarker inputs before constructing the mortality-associated physiological score:

$$X_{ij} \rightarrow R_{ij} \rightarrow \text{physiological axes} \rightarrow P_i$$

Because residualization and multivariable risk modeling do not generally commute, these procedures need not yield the same signal even when both final scores have little linear correlation with chronological age. PhenoAge NCP acceleration therefore provides a useful physiology-oriented comparator, but it is not mathematically identical to the PhysioRisk construction (Liu et al. 2018).

The framework is motivated by long-standing evidence of mortality heterogeneity among individuals of the same chronological age (Duchateau and Janssen 2007; Rockwood and Mitnitski 2007; Vaupel et al. 1979) and by work describing aging as multisystem physiological dysregulation (Cohen et al. 2013, 2015; Milot et al. 2014). PhysioRisk makes part of that heterogeneity observable through routine clinical biomarkers. PhysioRisk therefore captures mortality-associated physiological variation conditional on age and sex. It should not be interpreted as a direct measure of frailty or biological age, nor as a causal estimate of aging.

Finally, the decomposition separates two evaluation objectives that are often conflated. DemoRisk and CalibratedTotalRisk can be judged primarily as mortality-risk predictors, where discrimination and calibration are central. PhysioRisk can additionally be judged by whether it retains mortality information while remaining weakly associated with chronological age. A successful decomposition therefore does not require PhysioRisk alone to match the discrimination of a total-risk model. Doing so would largely defeat the purpose of removing demographic mortality structure. Instead, the relevant question is whether the residual physiological component is prognostically informative and whether recombining it with demographic risk adds information without obscuring the two sources of risk.

Methods

Study population, mortality linkage, and temporal split

Data came from nine continuous NHANES cycles from 1999–2000 through 2015–2016 (National Center for Health Statistics 1999–2016). Laboratory measurements were harmonized across survey cycles to prespecified units before biomarker construction. The source-cycle union contained 92,062 participants. Records were linked one-to-one by participant identifier to the 2019 public-use linked mortality files (National Center for Health Statistics 2022). Mortality eligibility required ELIGSTAT=1 and nonmissing follow-up time (PERMTH_INT) and mortality status (MORTSTAT), yielding 53,255 participants and 9,104 deaths. The endpoint was all-cause mortality. Follow-up began at the NHANES examination and ended at death or administrative censoring as recorded in PERMTH_INT. Participants without a recorded death were censored at their recorded follow-up month.

Complete data for the variables required by the prespecified biomarker construction were available for 19,829 participants (2,936 deaths). The downstream eight-biomarker no-CRP comparison removed no additional participants. Earlier cycles (1999–2000 through 2009–2010) formed the development cohort (n=13,090, 2,520 deaths). Later cycles

(2011–2012 through 2015–2016) formed the internal temporal-validation cohort (n=6,739, 416 deaths) (Figure 1 and Table 1).

NHANES uses a national probability sampling design, but analyses were unweighted because the objective was prediction-model development and temporal evaluation rather than population estimation. Estimates therefore describe this selected analytic cohort and are not population-representative NHANES estimates.

NHANES protocols were approved by the NCHS Research Ethics Review Board, and participants provided written informed consent at the time of data collection. The present analysis was a secondary analysis of de-identified public-use data and did not require additional institutional review board approval.

Missing data and cohort selection

For each cycle, missingness in the eight unique biomarkers required by PhysioRisk or PhenoAge NCP was quantified among mortality-eligible participants before complete-case restriction. Participants in the analytic cohort were compared descriptively with the 33,426 mortality-eligible participants excluded for missing one or more required fields. These comparisons summarized age, sex, deaths, and follow-up and did not modify the model specification (Supplementary Tables S6 and S7).

All five scores were evaluated in the same 6,739 complete temporal-validation participants. Training-median imputation for the PhenoAge NCP implementation consequently changed no values in this cohort. Common complete-case and common training-derived-imputation sensitivity analyses were recorded. Extension to excluded participants was not attempted because doing so would require a separately specified missing-data strategy and would change the retained complete-case validation design (Supplementary Table S8).

DemoRisk

DemoRisk was the log partial-hazard prediction from a Cox proportional hazards model (Cox 1972; Kleinbaum and Klein 2012) containing natural cubic spline terms for age (four degrees of freedom) and sex, with an L2 penalizer of 0.01. The model and spline design were fitted only in the 1999–2010 development cohort and applied unchanged to 2011–2016.

PhysioRisk

Each biomarker was regressed on natural cubic spline terms for age and sex in the development cohort. The fitted regression functions were applied to temporal-validation observations, and residuals were standardized using development-cohort means and standard deviations. HematologicStress was the signed average of standardized residuals for white blood cell count, lymphocyte percentage (reversed), mean corpuscular volume, and red cell distribution width. RenalHepaticTurnover similarly combined creatinine, albumin (reversed), and alkaline phosphatase.

The final two-axis score was applied unchanged in temporal validation. The model-development record does not establish that every decision leading to selection of the two axes was made without reference to later-cycle outcomes. The raw linear predictor used coefficients 0.419×HematologicStress + 0.239×RenalHepaticTurnover. The displayed coefficients are rounded. PhysioRisk was this raw predictor standardized using development-cohort parameters. Residualization functions, axis definitions, score coefficients, and standardization parameters were not refitted in temporal validation. Cox inference for PhysioRisk used a convergence-controlled fit and expressed the HR per one observed standard deviation in the temporal-validation cohort.

Calibrated total risk

The primary composite was defined as:

$$\mathrm{CalibratedTotalRisk} = 1.0650 \times \mathrm{DemoRisk} + 0.3610 \times \mathrm{PhysioRisk}$$

Both coefficients were estimated jointly in an unpenalized Cox model (Cox 1972; Kleinbaum and Klein 2012) using only the development cohort and were then fixed for temporal validation. Neither coefficient was selected using temporal-validation performance. The displayed coefficients are rounded to four decimal places. DemoRisk and PhysioRisk were also retained and evaluated separately.

Comparator

PhenoAge NCP was constructed from the published PhenoAge biomarker coefficients (Liu et al. 2018) with C-reactive protein omitted. Before applying the coefficients, albumin was converted to g/L, creatinine to μmol/L, and glucose was harmonized across cycles to mmol/L and entered linearly; the alkaline-phosphatase coefficient was 0.00188. It is therefore a PhenoAge no-CRP proxy, not the complete published implementation. PhenoAge NCP acceleration was the residual from a regression of PhenoAge NCP on spline-transformed age and sex fitted in development and applied unchanged to temporal validation.

Validation and statistical analysis

Known limitations of concordance measures for censored survival outcomes informed interpretation (Hartman et al. 2023). Kaplan–Meier estimation was used for observed risks and censoring distributions (Bewick et al. 2004).

All five primary scores were evaluated in the same temporal-validation cohort. Harrell C-index quantified rank discrimination (Harrell et al. 1996). Participant-level bootstrap resampling with 2,000 replicates (seed 20260821) provided percentile 95% confidence intervals for each C-index and paired confidence intervals for prespecified C-index contrasts. Cox HRs and 95% confidence intervals were estimated after division by each score's temporal-cohort sample standard deviation. Fits were checked for convergence. Pearson correlation summarized association with chronological age.

The proportional-hazards assumption was evaluated with rank-time Schoenfeld tests and scaled Schoenfeld residuals (Grambsch and Therneau 1994). A prespecified descriptive flag used $p<0.01$ without multiplicity adjustment. Separate diagnostic extended Cox models included standardized score × log(time/60 months) interactions as related time-varying-effect diagnostics (Grambsch and Therneau 1994). These temporal-cohort fits assessed time variation only and neither replaced nor recalibrated the development-derived prediction model.

Absolute mortality probabilities at 5 and 8 years were calculated from the development-cohort Cox coefficients and baseline survival. Calibration assessment followed established prognostic-model principles (Crowson et al. 2016; Royston 2014) and included mean predicted risk, Kaplan–Meier observed risk, observed-to-expected ratio, a global Cox calibration slope, decile calibration plots, and horizon-specific logistic calibration. For a horizon t, the binary outcome indicated death by t. Participants with known status at t contributed with inverse-probability-of-censoring weights derived from the temporal-validation censoring Kaplan–Meier estimate; participants censored before t had unknown binary status at the horizon and did not contribute as observed outcomes. A weighted logistic regression of the binary outcome on the logit of the fixed predicted risk estimated the calibration intercept and slope. The same IPCW scheme was used for the Brier score at each horizon (Graf et al. 1999). These references support the general survival-calibration and censoring-adjustment principles rather than defining this exact implementation. Temporal outcomes supplied evaluation weights and outcomes only. Prediction coefficients and baseline survival remained development-derived, and no temporal recalibration occurred.

As censoring-robust discrimination checks, Uno IPCW C truncated at 5 and 8 years (Uno et al. 2011) and cumulative/dynamic AUC (Heagerty et al. 2000) were calculated using the development cohort to estimate the censoring distribution. Follow-up support was summarized with reverse Kaplan–Meier follow-up and numbers at risk. These analyses were diagnostics of the fixed predictions rather than model-training steps.

Use of generative AI

OpenAI ChatGPT was used as an assistive tool during code development and troubleshooting, manuscript drafting and editing, and preparation of reproducibility and submission materials. The author reviewed and verified the analyses, numerical results, references, scientific interpretations, and final manuscript and assumes full responsibility for the submitted work.

Results

Cohort construction and selection

The cohort flow was 92,062 source-cycle participants, 53,255 mortality-eligible participants (9,104 deaths), and 19,829 biomarker-complete participants (2,936 deaths). The analytic cohort split into 13,090 development participants (2,520 deaths) and 6,739 temporal-validation participants (416 deaths) (Figure 1 and Table 1).

Among mortality-eligible participants, included participants had mean age 47.0 years (SD 19.1), were 50.6% female, had 2,936 deaths (14.8%), and had median follow-up of 124 months. The 33,426 excluded participants had mean age 47.8 years (SD 19.9), were 52.5% female, had 6,168 deaths (18.5%), and had median follow-up of 122 months. Complete-case selection therefore retained a group with a lower observed death proportion, limiting generalization beyond the analytic cohort (Supplementary Table S7).

Temporal-validation reverse Kaplan–Meier median follow-up was 74 months. There were 4,461 participants at risk at 5 years and 1,137 at 8 years, so the 8-year evaluation was supported by a smaller and less precise tail.

Discrimination, mortality association, and age correlation

DemoRisk provided most of the discrimination attributable to the model components, with C-index 0.827 (bootstrap 95% CI 0.804–0.847), HR 3.736 per temporal-cohort SD (95% CI 3.347–4.170), and age correlation 0.970 (Table 2). PhysioRisk, by contrast, remained only weakly correlated with age (r=0.102) but retained a mortality association, with C-index 0.663 (95% CI 0.632–0.694) and HR 1.385 (95% CI 1.335–1.436). Thus removing age-associated biomarker structure reduced standalone discrimination without eliminating prognostic information.

Recombining the two components improved overall risk ranking. CalibratedTotalRisk had C-index 0.844 (95% CI 0.823–0.864), HR 3.270 (95% CI 3.045–3.512), and age correlation 0.942. It exceeded DemoRisk by ΔC=0.0176 (paired 95% CI 0.0120–0.0237), indicating a small but supported increment from the physiological component.

PhenoAge NCP had C-index 0.845 (95% CI 0.823–0.864), HR 2.310 (95% CI 2.205–2.419), and age correlation 0.923. The difference between CalibratedTotalRisk and PhenoAge NCP was ΔC=−0.0003 (paired 95% CI −0.0073 to 0.0063). The temporal data therefore did not establish a discrimination difference between the two total-risk scores.

PhenoAge NCP acceleration had C-index 0.654 (95% CI 0.622–0.685), HR 1.298 (95% CI 1.256–1.341), and age correlation 0.051. PhysioRisk exceeded its C-index by 0.0088, but the paired 95% CI included zero (−0.0116 to 0.0289). The physiology-oriented comparison therefore favored PhysioRisk numerically without establishing a difference.

Fixed-horizon calibration

At 5 years, mean predicted mortality from CalibratedTotalRisk was 4.85% and Kaplan–Meier observed mortality was 4.65% (O/E 0.960). At 8 years, the corresponding values were 8.93% and 8.83% (O/E 0.988). Horizon-specific calibration slopes were 0.895 and 0.963, with intercepts −0.267 and −0.060 (Figure 2). These results show close agreement in average risk at both horizons, although the horizon-specific estimates did not have bootstrap confidence intervals. IPCW Brier scores were 0.0377 at 5 years and 0.0604 at 8 years.

The global Cox calibration slope was 0.804 (95% CI 0.755–0.852). Because this summary assumes a constant score effect, it cannot be interpreted simply as overfitting in the presence of the time-varying association reported below.

Proportional-hazards diagnostics

CalibratedTotalRisk violated the strict proportional-hazards assumption (rank-time Schoenfeld statistic 11.57, p=0.000670). A diagnostic score × log-time model gave an interaction coefficient of 0.130 (95% CI 0.029–0.230, p=0.01135). The positive estimate was compatible with an association that strengthened over follow-up, although the interaction did not meet the prespecified p<0.01 descriptive flag. This diagnostic model did not replace the fixed predictor.

Censoring-robust and missing-data sensitivity analyses

At 5 and 8 years, Uno C was 0.841 and 0.844 for CalibratedTotalRisk, 0.823 and 0.827 for DemoRisk, and 0.841 and 0.845 for PhenoAge NCP. Corresponding cumulative/dynamic AUCs were 0.851 and 0.870, 0.834 and 0.853, and 0.853 and 0.875. PhysioRisk Uno C was 0.666 and 0.663. The ordering and overall interpretation were therefore unchanged under censoring-robust metrics.

All 6,739 temporal-validation participants had the biomarkers required for both primary scores. Common complete-case analysis therefore reproduced PhysioRisk and PhenoAge NCP C-indices of 0.663 and 0.845, and common training-derived imputation changed no observations. Cycle-specific preselection missingness and the included/excluded comparison are reported in Supplementary Tables S6–S8.

Discussion

Principal findings

We evaluated whether mortality risk could be represented as two related but distinguishable quantities: demographic mortality structure associated with age and sex, and physiological variation remaining after biomarkers were adjusted for those variables. DemoRisk captured most of the overall mortality ranking. PhysioRisk had lower standalone discrimination but remained only weakly correlated with age and retained a mortality association. Recombining the components with development-estimated coefficients improved discrimination over DemoRisk alone and yielded performance similar to the PhenoAge no-CRP proxy.

The central finding is that measurable mortality-associated physiological heterogeneity remained after age- and sex-associated biomarker structure was removed. That component added prognostic information when returned to demographic risk while remaining separately observable. This supports the decomposition as an interpretable representation of mortality risk rather than evidence that PhysioRisk is a superior mortality predictor.

Low correlation with chronological age demonstrates statistical separation from age, not independence from biological aging. PhysioRisk should therefore be understood as a summary of mortality-associated physiological deviation conditional on age and sex rather than a direct measure of biological age.

Interpreting demographic and physiological risk separately

Chronological age is one of the strongest predictors of mortality, so removing it simply to make a score less age-correlated would discard useful information. The interpretive problem arises when total mortality risk and physiological condition are treated as the same quantity. DemoRisk describes position on the mortality gradient associated with age and sex, whereas PhysioRisk asks whether measured physiology is more or less adverse than expected for someone of the same age and sex. The components can therefore be examined separately or recombined when overall mortality prediction is the objective.

The lower standalone discrimination of PhysioRisk is expected because residualization intentionally removes age-associated biomarker structure that is highly predictive of mortality. Its relevant test is whether this residual physiological information remains associated with mortality and adds information when returned to demographic risk. Both were observed here.

More generally, total mortality prediction and characterization of physiological deviation are related but distinct objectives. Age correlation, component-specific discrimination, mortality association, and incremental performance after recombination therefore provide complementary information about the source of a score's predictive signal.

Relationship to biological-age and acceleration approaches

PhysioRisk differs from age-acceleration approaches in the stage at which adjustment occurs. Acceleration measures remove chronological-age structure from an already assembled biological-age or mortality score, whereas PhysioRisk removes age- and sex-associated structure from individual physiological measurements before they are combined and related to mortality. Because residualization and multivariable risk modeling are not generally interchangeable, the resulting signals need not be the same (Ferrucci et al. 2018; Klemera and Doubal 2006; Liu et al. 2018; Lu et al. 2019).

The comparison with PhenoAge NCP acceleration was consistent with this distinction. PhysioRisk had slightly higher discrimination, but the paired confidence interval included zero. The present data therefore do not establish that PhysioRisk is a stronger physiology-only mortality predictor.

The total-risk comparison was likewise neutral. CalibratedTotalRisk and PhenoAge NCP had similar discrimination, so decomposition did not obviously reduce risk ranking but did not improve it either. This comparison remains limited because PhenoAge NCP omits CRP and is not the complete published PhenoAge implementation (Liu et al. 2018).

Physiological axes and interpretability

The two retained physiological axes provide another level at which the model can be inspected. HematologicStress summarizes residual variation in white blood cell count, lymphocyte percentage, mean corpuscular volume, and red cell distribution width. RenalHepaticTurnover summarizes creatinine, albumin, and alkaline phosphatase after age- and sex-associated trends have been removed.

The labels are descriptive summaries of correlated clinical measurements, not claims of discrete biological systems or causal mechanisms. Their constituent biomarkers span hematopoietic, inflammatory, erythrocyte, renal, protein-homeostasis, and hepatobiliary processes. Aggregating them makes coordinated physiological variation inspectable without assigning causality to particular organ systems.

That interpretation is consistent with the physiological-dysregulation literature, which emphasizes coordinated disturbance across systems (Cohen et al. 2013, 2015; Milot et al. 2014). PhysioRisk approaches the same broad problem through survival modeling, asking which age- and sex-adjusted physiological deviations retain mortality information. The two-axis structure should therefore be regarded as a parsimonious implementation of the decomposition framework rather than a definitive taxonomy of aging physiology.

Recombination, calibration, and time dependence

Because DemoRisk and PhysioRisk occupy different numerical scales, their relative contributions to overall mortality risk were estimated jointly in the development cohort. The resulting combination was then applied unchanged in temporal validation.

In temporal validation, the physiological component provided a small but supported increment in discrimination over DemoRisk alone, while CalibratedTotalRisk remained similar to PhenoAge NCP. Thus the decomposition preserved overall risk ranking while keeping demographic and physiological contributions separately observable.

Fixed-horizon calibration remained close in the temporal cohort at both 5 and 8 years, and IPCW Brier scores and censoring-robust discrimination gave conclusions similar to conventional concordance. The 8-year estimate was less precise because fewer participants remained at risk (Crowson et al. 2016; Graf et al. 1999; Heagerty et al. 2000; Royston 2014; Uno et al. 2011).

CalibratedTotalRisk nevertheless violated the strict proportional-hazards assumption. The Schoenfeld diagnostic indicated nonproportionality, and the positive log-time interaction estimate was compatible with a mortality association that strengthened over follow-up, although the interaction did not meet the prespecified $p<0.01$ descriptive flag (Grambsch and Therneau 1994). This limits interpretation of a single constant hazard ratio and means the global Cox calibration slope should not be read simply as overfitting. Reasonable fixed-horizon prediction can coexist with a relative hazard association that changes with time.

The mechanism underlying that time dependence cannot be determined here. Future validation should examine time-specific performance rather than assume that a single proportional-hazards summary fully characterizes the composite.

Implications

Interpretability in mortality modeling need not require discarding chronological age, nor does strong prediction require demographic and physiological information to be collapsed into an inseparable score. For prognosis, the combined score is generally the relevant object. For questions about physiological heterogeneity among people of similar age and sex, the residual physiological component may be more informative even though its standalone discrimination is lower. In this setting, PhysioRisk prevents chronological age from answering a physiological question by proxy.

One potentially important application is longitudinal response assessment. Because PhysioRisk summarizes mortality-associated physiological deviation conditional on age and sex, repeated measurements could provide an intermediate readout before hard clinical outcomes accrue. At the individual level, change in PhysioRisk could provide a starting or interim measure of response. At the population level, shifts in its distribution could help characterize response heterogeneity. These uses require longitudinal validation of within-person reliability, responsiveness, and the relationship between change in PhysioRisk and subsequent outcomes.

The results also suggest a reporting principle for mortality-trained aging measures. Discrimination alone does not reveal whether a model's signal is predominantly demographic, physiological, or both. Reporting age dependence, mortality association, calibration, and incremental performance can make these contributions more explicit.

Limitations

The principal limitation is that the 2011–2016 evaluation is an internal temporal validation within NHANES rather than external validation in an independent population. Later survey cycles provide a meaningful test across calendar time, but they remain within the same broad survey, laboratory, and mortality-linkage ecosystem. Independent external validation is therefore required before claims of transportability or clinical use can be made.

The analysis also used an unweighted complete-case cohort rather than the full mortality-eligible NHANES sample. Of 53,255 mortality-eligible participants, 19,829 had the complete biomarker information required for the benchmark analysis. The 33,426 excluded participants had a higher observed death proportion than included participants, and biomarker missingness varied across survey cycles. The reported estimates therefore characterize the analytic cohort and should not be interpreted as population-representative NHANES estimates.

The benchmark comparator was a PhenoAge no-CRP proxy rather than the complete published implementation, so the comparison cannot establish superiority or equivalence to full PhenoAge. The physiological axes are statistical summaries of a limited panel of routine biomarkers, and other biomarker sets or modeling choices could capture different dimensions of physiological heterogeneity. The model-development record does not establish that every decision leading to the final two-axis specification was made without reference to later-cycle outcomes, so the temporal cohort cannot be regarded as a pristine never-examined holdout for every aspect of model selection. The final scoring specification is reproducible, and the CalibratedTotalRisk coefficients were estimated in development only.

Low correlation with chronological age demonstrates successful statistical adjustment, not causal separation from aging. Residualized biomarkers may continue to reflect age-related biological processes, disease burden, medication use, environmental exposures, socioeconomic factors, cohort effects, and measurement variation not represented by chronological age and sex alone.

Finally, calibration uncertainty was not bootstrapped at the individual fixed horizons, proportional hazards did not hold strictly for CalibratedTotalRisk, subgroup performance and clinical decision utility were not evaluated, and the 8-year estimates were supported by a relatively small tail of the temporal cohort. Participants excluded from the locked benchmark were not scored because extension beyond the retained complete-case cohort would require a separately specified missing-data strategy. These issues should be addressed alongside external validation before the framework is considered for clinical prediction.

Conclusions

Mortality risk associated with demographic position need not be represented by the same model component as mortality risk associated with how measured physiology differs from demographic expectation. Within NHANES, age- and sex-adjusted physiological variation retained mortality information, contributed incremental discrimination beyond demographic risk, and could be recombined with DemoRisk to achieve overall discrimination comparable to the no-CRP PhenoAge proxy.

PhysioRisk should therefore be viewed as an interpretable physiological-risk component. Its value lies in making demographic and physiological contributions separately observable while retaining the option to recombine them for total-risk prediction. It is not a replacement for chronological age or a definitive biological-age measure. Whether this decomposition transports beyond NHANES remains the principal question for external validation.

Statements and Declarations

Funding

This research received no external funding.

Ethics approval and consent to participate

This study used publicly available, de-identified NHANES data and public-use linked mortality files from the National Center for Health Statistics. NHANES protocols were approved by the NCHS Research Ethics Review Board, and participants provided written informed consent at the time of data collection. The present analysis was a secondary analysis of de-identified public-use data and did not require additional institutional review board approval.

Consent for publication

Not applicable.

Competing interests

The author declares no competing interests.

Author contributions

Gregory S. Allen conceived and designed the study, performed the analyses, interpreted the results, prepared the figures and tables, and wrote and revised the manuscript.

Data and code availability

NHANES 1999–2016 datasets used in this study are publicly available from the National Center for Health Statistics, Centers for Disease Control and Prevention, at https://www.cdc.gov/nchs/nhanes/. The 2019 Public-use Linked Mortality Files are publicly available from the National Center for Health Statistics at https://www.cdc.gov/nchs/data/datalinkage/public-use-linked-mortality-file-description.pdf. These public-use datasets do not have study-specific accession numbers or DOIs. Analysis code, aggregate result tables, supplementary material, and reproducibility and provenance documentation are available at https://github.com/allengr220/physiorisk_public/. Participant-level analytic datasets and derived score files are not redistributed. Their construction and provenance are documented in the repository.

Use of generative AI

OpenAI ChatGPT was used as an assistive tool during code development and troubleshooting, manuscript drafting and editing, and preparation of reproducibility and submission materials. The author reviewed and verified the analyses, numerical results, references, scientific interpretations, and final manuscript and assumes full responsibility for the submitted work.

Preprint

A previous version of this manuscript is available on Research Square. The current revised preprint is: Allen GS. PhysioRisk: an interpretable survival-modeling framework for separating demographic and physiological mortality risk. Research Square, Version 2, posted October 6, 2026. https://doi.org/10.21203/rs.3.rs-9829732/v2

Tables and figures

Table 1. Characteristics of the development and temporal-validation cohorts. Percentages use cohort size as the denominator. Follow-up is reported in months. Detailed source-cycle counts and the mortality-eligible included/excluded comparison are provided in Supplementary Tables S6 and S7.

Characteristic	Development (1999–2010)	Temporal validation (2011–2016)
Participants	13,090	6,739
Age, mean (SD), years	46.7 (19.5)	47.6 (18.2)
Female, n (%)	6,652 (50.8%)	3,390 (50.3%)
Deaths, n (%)	2,520 (19.3%)	416 (6.2%)
Follow-up, median (IQR), months	159 (125–203)	71 (53–90)

Table 2. Model performance in the temporal-validation cohort. All models were evaluated in 6,739 participants with 416 deaths. C-index intervals are percentile intervals from 2,000 participant bootstrap replicates. Hazard ratios are per temporal-validation-cohort standard deviation. Paired intervals for direct C-index contrasts are reported in the Results and Supplementary Table S1.

Model	C-index (95% CI)	HR per validation SD (95% CI)	Age correlation
DemoRisk	0.827 (0.804–0.847)	3.736 (3.347–4.170)	0.970
PhysioRisk	0.663 (0.632–0.694)	1.385 (1.335–1.436)	0.102
CalibratedTotalRisk	0.844 (0.823–0.864)	3.270 (3.045–3.512)	0.942
PhenoAge NCP	0.845 (0.823–0.864)	2.310 (2.205–2.419)	0.923
PhenoAge NCP acceleration	0.654 (0.622–0.685)	1.298 (1.256–1.341)	0.051

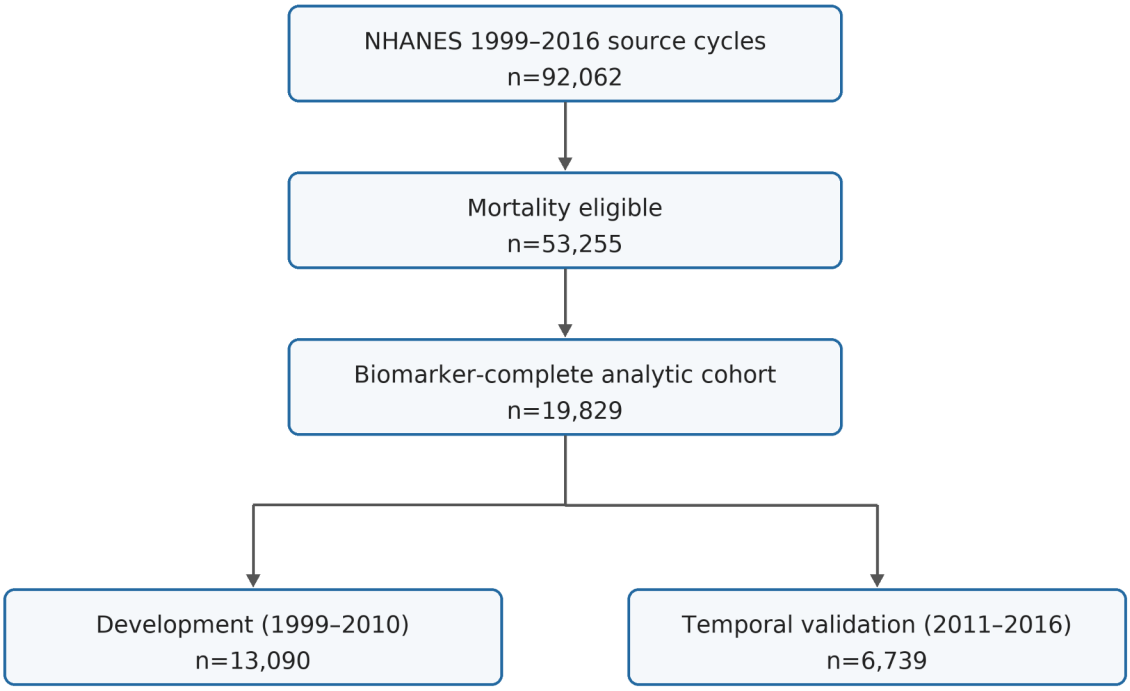


Figure 1. Cohort flow. Flow of NHANES 1999–2016 participants from the source-cycle population through mortality eligibility and biomarker completeness to the development and internal temporal-validation cohorts. Counts of deaths are reported in the text and Table 1.

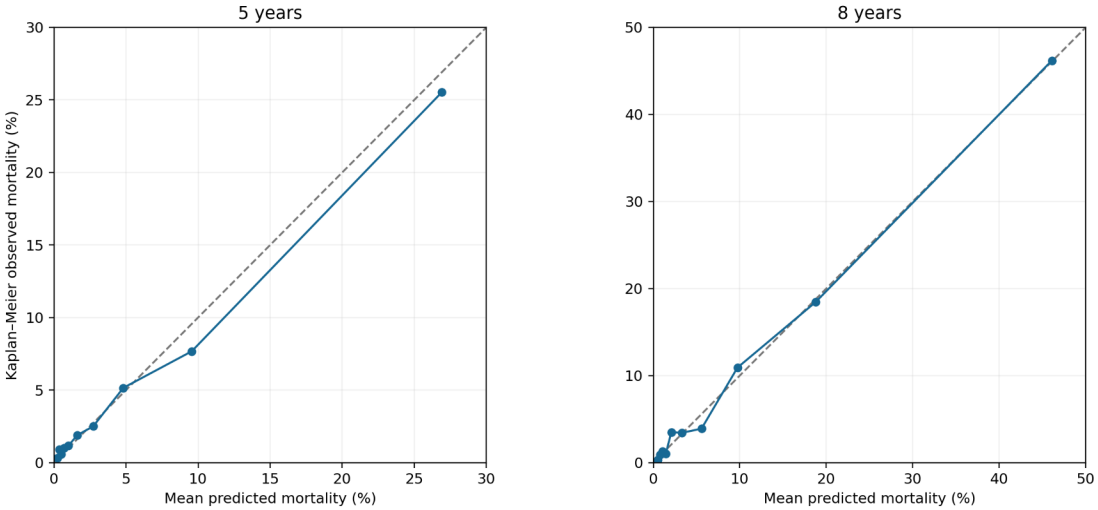


Figure 2. Fixed-horizon calibration of CalibratedTotalRisk in temporal validation. Mean predicted mortality and Kaplan–Meier observed mortality are shown by risk decile at 5 and 8 years. The dashed diagonal denotes perfect agreement.

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